

Joel Sam Mathew

github.com/joelmathew003

linkedin.com/in/joel-mathew-3122751a6/

Munich, Germany

✉: joelsammathew@gmail.com

☎ +49 15510942045

EDUCATION

Technical University of Munich (TUM)

M.Sc. Informatics – Incoming

Munich, Germany

October 2026

Indian Institute of Technology (IIT) Palakkad

B.Tech. Computer Science and Engineering, CGPA: 9.02/10

Palakkad, India

August 2019 – May 2023

WORK EXPERIENCE

Arista Networks

Software Engineer

Bangalore, India

July 2025 – September 2026

- Closed a supply-chain trust gap in EOS VM image releases by implementing CMS-based cryptographic signing, giving customers a way to verify authenticity that hadn't existed before. Published a customer-facing **TOI** for signature verification.
- Closed an unaudited access path into the release repository — traced ~100K daily Artifactory calls (at peak) under one shared token back to their sources, then migrated each to per-identity Dex/OIDC authentication.
- Audited the WiFi product team's fully manual release process and engineered an automated pipeline integrated with the EOS release framework.
- Owned operational health of the release build server fleet, which includes reimaging, deploying release-related services to production, building Grafana dashboards tracking server health and rollouts etc. Also setup SSH audit logging across release build servers, with events forwarded to CrowdStrike Falcon SIEM for centralized monitoring.

ColorTokens Inc.

Member of Technical Staff – II

Bangalore, India

July 2023 – July 2025

- Developed and integrated a dynamic rule engine for policy enforcement within XShield, a zero-trust management platform, reducing infrastructure overhead for key components by over 50% while improving scalability.
- Designed and implemented Layer 3 network policy enforcement within the container micro-segmentation framework using Istio and OPA, enabling fine-grained control over network traffic paths.
- Implemented an opt-in north-south traffic visibility feature, a platform-level switch enabling packet telemetry collection on the server agent, improving control over traffic monitoring and analytics.
- Sole technical point of contact for container micro-segmentation across customer POCs, resolving production issues end to end.

SELECTED PROJECTS

PlexShare

Aug – Dec 2022

- Contributed to the development of a lab session monitoring app in C#, involving features like screensharing, collaborative whiteboard, file uploads, and chat functionalities. Implemented key functionalities in the whiteboard module such as session loading and saving, data serialization and storage, and inter-module networking.
- Stress-tested with a full classroom (~30 concurrent users) screen-sharing and whiteboarding simultaneously in a live trial.

Mail Tag Generator

Jan – May 2023

- Designed and developed a personalized email tagging system combining Latent Dirichlet Allocation (LDA) for topic modeling with GloVe embeddings for candidate tag ranking.
- Implemented the system as a Chrome extension in JavaScript, generating contextually relevant tags for Gmail messages in real time.

Android Malware Detection using GNN

Jan – May 2022

- Developed an Android malware detection approach using Graph Neural Networks trained on API call graphs.
- Applied explainability techniques including GNNExplainer and SubgraphX to identify influential subgraphs and investigate model predictions.

Tiger Compiler

Jan – May 2022

- Developed a compiler for the Tiger programming language in Standard ML using ML-Lex and ML-Yacc, generating MIPS assembly code.

TECHNICAL SKILLS

Languages: Go, Python, C++, C#

Infrastructure & Tools: Kubernetes, Docker, Linux, Ansible, Grafana, Artifactory, CI/CD, Azure Pipelines

AI/ML: TensorFlow, PyTorch

ACHIEVEMENTS

- ColorKudos Award — Individual, for contributions to the Port-Level Zero Trust Enforcement project (ColorTokens, 2024)
- ColorKudos Award — Team, for backend contributions to XShield's policy engine (ColorTokens, 2024)
- Scored 99.67 percentile in JEE Main (1.2M candidates) and 98.11 percentile in JEE Advanced (2019)
- Qualified at the state level for Kerala's Young Innovators Program for an autonomous pesticide-spraying robot (2019)